Ex:10:- What is Avg of 1st 12 multiples of 8?

Ans:
(M-1) $Avg = \frac{1\times8+2\times8+3\times8 \cdots 12\times8}{12} = \frac{8[1+2+\cdots 12]}{12}$

$= 8[\text{Avg of 1st 12 natural Nos}] = 8\left[\frac{12+1}{2}\right]$

$= 4\times13 = 52$

M-2:- All are in A.P $[8, 16, 24, \cdots 12\times8=96]$

$\therefore Avg = \frac{1^{st}\text{ obs} + \text{last obs}}{2} = \frac{8+96}{2} = \frac{104}{2} = 52$

M-3:- Whenever all given observations are in A.P & even no. of observations given, then Avg is Average of middle 2 observations. In 12 observations 6th & 7th are middle 2.

$Avg = \frac{6^{th}\text{obs} + 7^{th}\text{obs}}{2} = \frac{6\times8+7\times8}{2}$

$= \frac{48+56}{2} = \frac{104}{2} = \underline{52}$

Ex:11:- Average weight of class of 50 students is 120 kgs. If teacher is added, the average is red by 3 kgs. The weight of the teacher is?

Ans:- (M-1) Sum of weights of 50 students $= 50\times120 = 6000$ kgs - (1)

Sum of wts of students & teacher $= (50+1(\text{teacher}) \times (120+3)$ red by 3

$= 51\times123 = 6273$ - (2)

weight of teacher = (2) - (1) $= 6273 - 6000 = \underline{273}$ kgs.

(M-2) By using definition:- Whatever extra 3 kgs distributed to all including teacher is retrieved back & added to common average to find weight of teacher.

Wt of teacher = Common Avg + 51 × 3 kgs = 120 + 153 = (273)

value of added. = Common Avg + (Original + added) red value.

Ex-12:- The avg age of 60 members is 25. If new people are joined, the avg ged by 2yrs then.

(i) What is Avg age of 2 new people.

(ii) What are their ages if differed by one year.

(iii) What are their ages if their ages ratio is 173:175?

Ans:- (i) Sum of Ages of 2 people = 2 C.A + (60+2) × 2

$= 50 + 124 = 174$ —— (1)

Avg of Ages of 2 people $= \frac{174}{2} = 87$ yrs

(ii) Lets assume younger one of 2 as 'x' then other 'x+1'

Ans $x + x + 1 = 174 \Rightarrow 2x = 173 \Rightarrow x = 86.5$ yrs

Ages = 86.5 yrs & 87.5 yrs

(iii) Ages ratio = 173:175

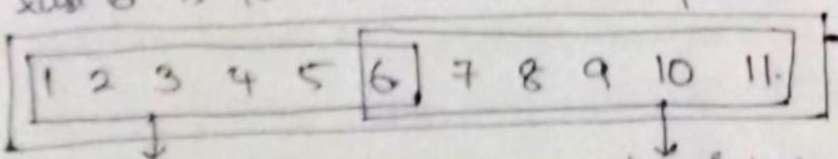
$\Rightarrow$ 1st Age = 173x, 2nd Age = 175x

from (1)

$173x + 175x = 174 \Rightarrow x = 0.5$

1st Age = 173 × 0.5 = 86.5 yrs, 2nd Age = 175 × 0.5 = 87.5 yrs.

Ex-13:- The Average score of cricket team is 45 runs in which, the average of first 6 members is 50 & that of last 6 is 43. The score of 6th person?



Total Avg = 45

Total score = 45 × 11 = 495

Avg of 1st 6 = 50

Total score of 1st 6 = 50 × 6 = 300

Avg of last 6 = 43

Total score of last 6 = 43 × 6 = 258

6th person score = 1st 6 sum + last 6 sum − Total score

[11 + 6th score] —— [11]

$= 300 + 258 - 495 = \boxed{63}$

(or)

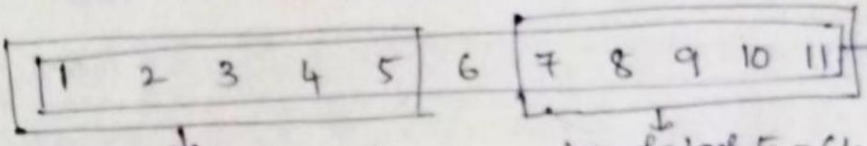
By using definition whatever distributed that is retrieved back to get required one value.

6th person score = C.A + (50-45)6 - (45-43)6

= 45 + 30 - 12 = 63

Ex:14:- The average score of cricket team is 60 and that of first 5 Avg is 57 & that of last 5 Avg is 64. The score of 6th person / batsman / player is ?

Ans:-



Avg of 1st 5 = 57
Sum of " " = 5x57
= 285

Avg of last 5 = 64
Sum " " = 64x5
= 320

Avg of 11 = 60
Sum = 60x11
= 660

6th player score = Total 11 score - (1st 5 sum + last 5 sum)

= 660 - (320 + 285) = 660 - 605

6th player score = 55

(or)

By using definition of distribution:-

6th person score = C.A + (60-57)5 - (64-60)5

= 60 + 15 - 20 = 55

Note 1:- When one person is coming extra like in Ex-13, then retreiving like as it is according to definition [↑ → +, ↓ → -]

Note 2:- When one person is skipped like in Ex:14, then retreiving is opposite scenario like [↑ → -, ↓ → +.]

Ages [Vector India Pvt. Ltd]

- By Sunil Guptha Yaramalla.

↳ In ages we must know few points that are very important are.

Ago / before / was ⟶ '−' to present Age

After / Next / In / hence / will ⟶ '+' to present Age

↳ In ages all concepts what we have discussed till now are used to do solutions. i.e., Application of concepts of %'s, Ratio's & Avg's on Ages only concept of Ages.

↳ Example of Sandeep present age as 'x' then

6 yrs Ago / before / ago age was ⟶ $x-6$

after / Next / In / hence 6 yrs / 6 yrs ahead will be ⟶ $x+6$

Ex:-15:- 20 yrs ago Sameer age was half of his present age. The present age of Sameer is?

Ans:- Let's Sameer present age = S = 40 yrs

20 yrs ago = $S-20$, $\frac{1}{2}$ of age = $\frac{S}{2}$

$\therefore S-20 = \frac{S}{2} \Rightarrow 2S-40 = S \Rightarrow S = 40$ yrs

(or)

20 yrs ago = 20 yrs = completed age & $\frac{1}{2}$ to total age.

$\Rightarrow 20 = \frac{S}{2} \Rightarrow S = 40$ yrs.